The Chinese University of Hong Kong, Shenzhen
SDS · School of Data Science

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MAT 3007 — Optimization

Exercise Sheet 8

Problem 1 (Non-Lischitz Smooth Function): (*20* pts)
In Assignment 7, we have shown $f(x) = |x|^{3/2}$ is a convex function.

a) Prove f is not Lipschitz smooth. (Recall the definition: if $f : \mathbb{R}^n \to \mathbb{R}$ is Lipschitz smooth with constant $L > 0$, then $\|\nabla f(\boldsymbol{x}) - \nabla f(\boldsymbol{y})\| \leq L\|\boldsymbol{x} - \boldsymbol{y}\|$ for any $\boldsymbol{x}, \boldsymbol{y}$.)

b) Prove that for any constant step size $\bar{\alpha}$ in gradient descent method applied to this function, there exists x_0 for which $f(x_1) \geq f(x_0)$.

Remark: This problem highlights the importance of Lipschitz smoothness when using constant step size in the gradient descent method.

Problem 2 (Non-Strongly Convex Function): (*20* pts)
Consider a function $f(x) = x^2 + 3\sin^2 x$.

a) Show that this function is not convex. Verify that $x^* = 0$ is a global minimizer.

b) Show that this function is Lipschitz smooth with some constant $L > 0$.

c) Consider the gradient descent method with a constant step size $\bar{\alpha}$. Show that after one iteration, the following inequality holds:

$$f(x_{k+1}) \leq f(x_k) - \left(\bar{\alpha} - \frac{L\bar{\alpha}^2}{2}\right)|f'(x_k)|^2.$$

You can directly use the following lemma.

Lemma 1 If g is a Lipschitz smooth function with constant L, then

$$g(\boldsymbol{y}) \leq g(\boldsymbol{x}) + \nabla g(\boldsymbol{x})^\top(\boldsymbol{y} - \boldsymbol{x}) + \frac{L}{2}\|\boldsymbol{y} - \boldsymbol{x}\|^2, \qquad (\star)$$

for any $\boldsymbol{x}$ and $\boldsymbol{y}$.

d) It can be shown that $|f'(x)|^2 \geq \frac{1}{16}(f(x) - f(x^*)), \forall x$ (you don't need to verify this). Combining this with part (c), show linear convergence in the following sense:

$$f(x_{k+1}) - f(x^*) \leq \eta[f(x_k) - f(x^*)].$$

for some $\eta \in (0, 1)$, with a proper step size $\bar{\alpha}$.

Remark: This problem shows that although the assumptions of Lipschitz smoothness and strong convexity are important for linear convergence, but they are not always necessary.

Problem 3 (Bisection and Golden Section Methods): (*20* pts)

Consider the optimization problem:

$$\min_x \ 4x^2 - x^4.$$

Starting from the interval $[-1, 1/2]$, solve the problem using both the Bisection Method and the Golden Section Method in MATLAB or Python. For both methods, output the midpoint of the final interval as the approximate solution. Compare the number of iterations required by each method to achieve a solution that is within 10^{-4} of the optimal solution in the given starting interval, $x^* = 0$.

Problem 4 (True or False): (*20* pts)

For each of the following statements, determine whether it is true or false. Provide a justification for your answer. Assume that we are minimizing a differentiable function f.

a) At the iterate $\boldsymbol{x}_k$, define a search direction $\boldsymbol{v}$ as follows:

$$\boldsymbol{v}_i = \begin{cases} -1 & \text{if the } i\text{-th coordinate of } \nabla f(\boldsymbol{x}_k) \text{ is positive,} \\ 1 & \text{if the } i\text{-th coordinate of } \nabla f(\boldsymbol{x}_k) \text{ is negative,} \\ 0 & \text{if the } i\text{-th coordinate of } \nabla f(\boldsymbol{x}_k) \text{ is zero.} \end{cases}$$

This search direction is a descent direction as long as $\boldsymbol{x}_k$ is not a stationary point.

b) In gradient descent with backtracking line search, it is always guaranteed that: $f(\boldsymbol{x}_{k+1}) < f(\boldsymbol{x}_k)$ as long as $\boldsymbol{x}_k$ is not a stationary point.

c) In gradient descent with exact line search, it is always guaranteed that: $f(\boldsymbol{x}_{k+1}) < f(\boldsymbol{x}_k)$ as long as $\boldsymbol{x}_k$ is not a stationary point.

d) Gradient descent will always converge to a local minimizer when using exact line search.

Problem 5 (Step Size Strategies): (*20* pts)

Consider the optimization problem:

$$\min_x f(\boldsymbol{x}) = x^\top \boldsymbol{A} x, \quad \text{where } \boldsymbol{A} = \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix}.$$

Starting from the initial point $\boldsymbol{x}_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, apply **gradient descent** using the update rule:

$$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k - \alpha_k \nabla f(\boldsymbol{x}_k)$$

with the following step size strategies:

a) Compute the following by hand or using a calculator:

- Perform two steps of gradient descent with a constant step size of 0.1, i.e., $\alpha_k = 0.1$ for all k.
- Perform one step of gradient descent with exact line search.

- Perform one step of gradient descent with backtracking line search: Search for α_k that satisfies the Armijo condition:

$$f\left(\boldsymbol{x}_k - \alpha \nabla f(\boldsymbol{x}_k)\right) \leq f\left(\boldsymbol{x}_k\right) - \gamma\alpha \cdot \|\nabla f\left(\boldsymbol{x}_k\right)\|^2$$

  If the condition is not satisfied, update $\alpha \leftarrow \sigma\alpha$. Use the parameters $\gamma = \frac{1}{2}$ and $\sigma = 0.2$.

b) Use Matlab or Python to implement the above three step size strategies and run gradient descent for 10 iterations. Provide their last iterates and objective function values.

Sheet 8 is due on **May 2nd**. Submit your solutions before **May 2nd**, **11:00 pm**.